JuMP Syntax Cheatsheet

Optimization with Julia



This cheatsheet summarizes the most common syntax elements of JuMP for optimization modeling in Julia. It includes examples for setting up models, declaring variables and constraints, defining objectives, adjusting solver options, solving, and accessing solutions.

> **Note:** This cheatsheet uses the latest JuMP syntax. For more detailed documentation, visit the JuMP documentation.

Model Setup

Importing Packages and Creating a Model

```
using JuMP, HiGHS

# Create a model with HiGHS optimizer
model = Model(HiGHS.Optimizer)

# Set a time limit for the optimizer (optional)
set_optimizer_attribute(model, "time_limit", 60.0)
```

Variables

Declaration

Continuous Variables

```
# Unbounded continuous variable
@variable(model, x)

# Non-negative continuous variable
@variable(model, x >= 0)

# Bounded continuous variable (0 <= x <= 10)
@variable(model, 0 <= x <= 10)

# Fixed variable, x is fixed at 5
@variable(model, x == 5)
```

Integer Variables

```
# Unbounded integer variable
@variable(model, x, Int)

# Non-negative integer variable
@variable(model, x >= 0, Int)

# Bounded integer variable (0 <= x <= 10)
@variable(model, 0 <= x <= 10, Int)
```

Binary Variables

```
# Binary variable (0 or 1)
@variable(model, x, Bin)
```

Containers

Arrays and Matrices

```
# Array of 5 continuous variables
@variable(model, x[1:5])

# Non-negative array of variables
@variable(model, x[1:5] >= 0)

# Binary array of variables
@variable(model, x[1:5], Bin)

# 3x4 matrix of variables
@variable(model, x[1:3, 1:4])

# Integer 3x4 matrix
@variable(model, x[1:3, 1:4], Int)
```

Custom Indexing

```
indices = ["A", "B", "C"]
@variable(model, x[i in indices])
```

Constraints

Basic Constraints

```
# Declare additional variables as needed
@variable(model, x)
```

```
@variable(model, y)

@constraint(model, con1, 2x + y <= 10)
@constraint(model, con2, x + 2y >= 5)
@constraint(model, con3, x == y)   # Equality constraint
```

Constraints with Containers

```
# Array Variables
@variable(model, x[1:5] >= 0)

# Constraint for each variable element
@constraint(model, capacity[i=1:5],
    x[i] <= 100
)

# Sum constraint across array
@constraint(model, total_sum,
    sum(x[i] for i in 1:5) <= 500
)

# Matrix constraints: each element gets its constraint
@variable(model, y[1:3, 1:4])
@constraint(model, matrix_con[i=1:3, j=1:4],
    y[i,j] <= i + j
)
```

Conditional Constraints

```
# Constraint applies only for indices where i > 2
@constraint(model, cond[i=1:5; i > 2],
    x[i] <= 10
)

# Multiple conditions for two-dimensional index:
@constraint(model, cond2[i=1:10, j=1:10; i != j && i + j <= 15],
    x[i,j] + x[j,i] <= 1
)
```

Constraints with Conditional Summations

```
# Sum over a subset of indices (e.g., i > 2)
@constraint(model, total_sum,
    sum(x[i] for i in 1:5 if i > 2) <= 500
)
```

```
# Sum with multiple conditions on a 2D array
@constraint(model, total_sum2,
    sum(x[i,j] for i in 1:5, j in 1:5 if i != j && i + j <= 7) <= 1
)
```

Tip: Use semicolons (`;`) to separate the index definition from conditions when defining constraints.

Objective Functions

Basic Objectives

```
@objective(model, Max, 5x + 3y)   # Maximize
@objective(model, Min, 2x + 4y)   # Minimize
```

Objectives Using Containers

```
# Container-based objective
@variable(model, z[1:10])
@objective(model, Min, sum(z[i] for i in 1:10))

# Weighted objective function
weights = [1, 2, 3, 4, 5]
@objective(model, Max,
    sum(weights[i] * z[i] for i in 1:5)
)
```

Solver Options

```
# Recreate model with a solver if needed
model = Model(HiGHS.Optimizer)

# Set a 60-second time limit
set_time_limit_sec(model, 60)
println("Current time limit: ", time_limit_sec(model))

# Tolerance attributes: relative and absolute gap
set_optimizer_attribute(model, "mip_rel_gap", 0.01)  # 1% gap tolerance
set_optimizer_attribute(model, "mip_abs_gap", 0.1)   # Absolute gap tolerance

# Presolve options: enable or disable
```

```
set_optimizer_attribute(model, "presolve", "on")  # Enable presolve
# set_optimizer_attribute(model, "presolve", "off")  # To disable presolve
```

Tip: Adjust these settings to balance between solution precision and computational speed.

Solving and Inspecting the Model

```
# Optimize the model
optimize!(model)

# Check the termination status using MathOptInterface statuses
status = termination_status(model)
if status == MOI.OPTIMAL
    println("Solution is optimal")
elseif status == MOI.TIME_LIMIT && has_values(model)
    println("Time limit reached with a feasible solution")
else
    println("Solver status: ", status)
end

# Get variable and objective values
x_val = value(x)
println("x value: ", x_val)

# For arrays, retrieve values with broadcasting:
x_vals = value.(x)
println("x array values: ", x_vals)

# Get the objective value
obj_val = objective_value(model)
println("Objective value: ", obj_val)
```

Important: Always check the termination status before using the solution values.

Model Modifications

Variable Updates

```
# Set or update variable bounds
set_lower_bound(x, 0)
set_upper_bound(x, 10)
```

```
# Fix x to a specific value, then unfix if needed
fix(x, 5)
unfix(x)
```

Constraint Updates

```
# Delete a constraint if necessary
delete(model, con1)
```

Note: After modifying the model, you will need to re-solve it to update the solution.

Additional Features and Advanced Topics

Checking Variable Properties

```
# Check bounds for a variable
println("Has lower bound? ", has_lower_bound(x))
println("Lower bound: ", lower_bound(x))
println("Has upper bound? ", has_upper_bound(x))
println("Upper bound: ", upper_bound(x))

# Check variable type and retrieve information
println("Is x binary? ", is_binary(x))
println("Is x integer? ", is_integer(x))
println("Variable name: ", name(x))
println("Total number of variables: ", num_variables(model))
```

Simple Model Example

```
# Create a simple model example
   using JuMP, HiGHS

   model = Model(HiGHS.Optimizer)
   @variable(model, 0 <= x <= 10)
   @constraint(model, con1, x >= 5)
   @objective(model, Max, x)

   optimize!(model)

   # Introspection on the variable x
```

```
    println("Has lower bound? ", has_lower_bound(x))
    println("Lower bound: ", lower_bound(x))
    println("Has upper bound? ", has_upper_bound(x))
    println("Upper bound: ", upper_bound(x))
    println("Is x binary? ", is_binary(x))
    println("Is x integer? ", is_integer(x))
    println("Is x fixed? ", is_fixed(x))
    println("Variable name: ", name(x))
    println("Total number of variables: ", num_variables(model))
```

```
Running HiGHS 1.12.0 (git hash: 755a8e027): Copyright (c) 2025 HiGHS under MIT
licence terms
LP has 1 row; 1 col; 1 nonzero
Coefficient ranges:
  Matrix  [1e+00, 1e+00]
  Cost    [1e+00, 1e+00]
  Bound   [1e+01, 1e+01]
  RHS     [5e+00, 5e+00]
Presolving model
0 rows, 0 cols, 0 nonzeros  0s
0 rows, 0 cols, 0 nonzeros  0s
Presolve reductions: rows 0(-1); columns 0(-1); nonzeros 0(-1) - Reduced to
empty
Performed postsolve
Solving the original LP from the solution after postsolve

Model status        : Optimal
Objective value     :  1.0000000000e+01
P-D objective error :  0.0000000000e+00
HiGHS run time      :          0.00
Has lower bound? true
Lower bound: 0.0
Has upper bound? true
Upper bound: 10.0
Is x binary? false
Is x integer? false
Is x fixed? false
Variable name: x
Total number of variables: 1
```

Key Points

- Always check solution status before using results
- Set appropriate time limits for large problems
- Use gap tolerances to balance precision and speed
- Monitor solve time for performance optimization
- Consider presolve for complex problems